

Similarity by Selected Traits

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Repair Bias in Data

Without opening the old lesson, explain one key idea from Repair Bias in Data. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Missing Data

Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Choose Informative Questions

Without opening the old lesson, explain one key idea from Choose Informative Questions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Compute distances for Distance between feature vectors. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Distance between feature vectors. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Rank independently

Target is [3, 2]. Options A [1, 2], B [3, 4], C [4, 2]. Calculate distances, select the closest, and identify a tie among the others.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Represent books as [animal,short,illustrated] with 1=yes. Query=[1,1,0]; A=[1,0,0], B=[1,1,1], C=[0,0,0]. Count mismatches. What changes if illustrations matter twice as much?

6. Solve the given case

Traits are [animals,adventure]. A=[1,1], B=[1,0], C=[0,0]; query=[1,0].

Use number of differing traits as distance.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Score similarity using one point for same topic and one for same difficulty. Target is space, easy. A is space, hard; B is animals, easy. What happens?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 21 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Make selected traits and their scales visible in every recommendation explanation.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Represent books as [animal,short,illustrated] with 1=yes. Query=[1,1,0]; A=[1,0,0], B=[1,1,1], C=[0,0,0]. Count mismatches. What changes if illustrations matter twice as much?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Read a vector

For [4, 1] under the worked feature order, identify energy and group-work ratings.

My reasoning and evidence

13. W2 Compute distances

From target [1, 2], find distances to A [2, 2] and B [0, 4].

My reasoning and evidence

14. W3 Repair feature order

One record uses [group work, energy] while others use [energy, group work]. Why is the distance unreliable?

My reasoning and evidence

15. W4 Rank independently

Target is [3, 2]. Options A [1, 2], B [3, 4], C [4, 2]. Calculate distances, select the closest, and identify a tie among the others.

My reasoning and evidence

16. W5 Limit similarity

Two activities have distance zero. Must they suit every learner equally well?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

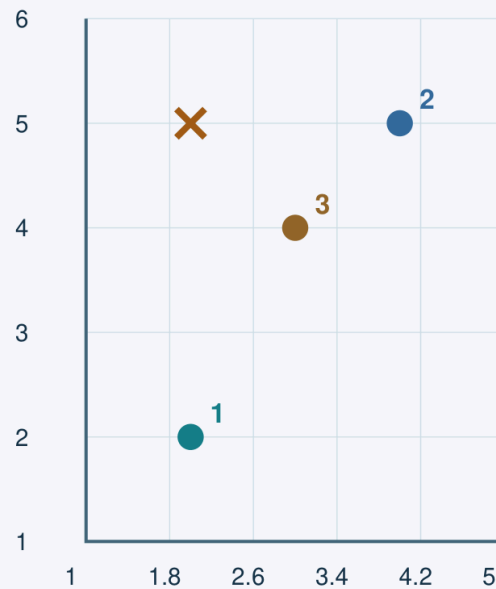
18. Pictorial case practice: Similarity by Selected Traits

A fictional toy book matcher represents two preferences with numeric features.

Target vector [2,5]. Candidate A=[2,2], B=[4,5], C=[3,4].

Use Manhattan distance: add absolute coordinate differences; return all smallest-distance candidates.

Toy feature positions



Horizontal: first toy feature; vertical: second toy feature

1: A at (2, 2)

2: B at (4, 5)

3: C at (3, 4)

Cross: Target at (2, 5)

Distance rule: Manhattan distance; horizontal=first feature, vertical=second feature

Similarity by Selected Traits

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Distance A

$$|2-2|+|5-2|=0+3=3.$$



2. Distance B

$$|2-4|+|5-5|=2+0=2.$$



3. Distance C

$$|2-3|+|5-4|=1+1=2.$$



4. Choose closest

B and C tie at distance 2; keep the tie visible.

CHANGE ONE THING

Give the first-coordinate difference weight 2 while the second keeps weight 1.

Explain the changed result using the matching lesson.

Explain why B ceases to be a closest option without any candidate data changing.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Specify recommender feature meanings, units, scaling and tie policy before explaining nearest matches.

19. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Specify recommender feature meanings, units, scaling and tie policy before explaining nearest matches.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional toy book matcher represents two preferences with numeric features.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? B and C tie at distance 2; keep the tie visible.

1. Distance C
2. Choose closest
3. Distance A
4. Distance B

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Choose closest B and C tie at distance 2; keep the tie visible. Each coordinate contributes according to the declared distance rule. Smaller distance means closer in this representation, not automatically a better reading experience. If one coordinate uses a much larger numerical scale, it can dominate the sum even when it is not more important. Feature units and scaling belong in the method record.

Practice 2: explain the deciding evidence

Changed-case practice: Give the first-coordinate difference weight 2 while the second keeps weight 1. Which conclusion is supported by the supplied facts?

1. Distances become $A=3$, $B=4$, $C=3$; A and C tie.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Distances become $A=3$, $B=4$, $C=3$; A and C tie. The representation's comparison rule changes; the supplied vectors stay the same.

Repair a missing building block

Review Repair Bias in Data (ai-2/20). Goal: Identify selection and coverage bias. Short practice: Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?. Return to this week and explain what you changed.

Review Missing Data (ai-2/18). Goal: Detect absent and unusable values. Short practice: Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?. Return to this week and explain what you changed.

Review Choose Informative Questions (ai-2/15). Goal: Fit a root question and leaf predictions from training examples. Short practice: Choose the best original question and explain why its more balanced split matters here.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
